

floorcall

A millisecond decision layer for voice agents. Project report.

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Code: <https://github.com/yashraz23/floorcall>

Model: <https://huggingface.co/enz23/floorcall>

Demo: <https://huggingface.co/spaces/enz23/floorcall>

Summary

A voice agent has to settle a few questions inline, before it acts: has the user finished, was the speech over the agent a backchannel or a real interruption, what does the user want, and should a person take over. floorcall answers them together, from the conversation's text, in one batched forward pass of Laya, a small open-weight decision encoder, fine-tuned on all four tasks and calibrated per decision. Every number below is on a frozen test set that no training, calibration or threshold choice saw.

- D1 turn_complete: macro-F1 0.813 [0.807, 0.820]; against the best cheap baseline (TF-IDF + logistic regression), +0.269 [+0.259, +0.279] paired.
- D2 barge_in: macro-F1 0.937 [0.931, 0.943]; against the best cheap baseline (a lexical rule), +0.029 [+0.022, +0.035] paired.
- D3 route: macro-F1 0.906 [0.881, 0.926]; against the best cheap baseline (TF-IDF + logistic regression), +0.058 [+0.032, +0.085] paired.
- D4 escalate: macro-F1 0.681 [0.612, 0.747]; against the best cheap baseline (TF-IDF + logistic regression), +0.000 [-0.061, +0.060] paired.
- Calibration: per-decision temperature takes ECE from 0.020 to 0.010 (D1), 0.008 to 0.003 (D2), 0.043 to 0.011 (D3), 0.277 to 0.064 (D4).
- Latency (laptop GPU, batch 1, the pause event's three questions in one call): p50 29.1 ms, p99 59.0 ms, against a 50 ms p99 budget that no measured path meets.
- Normalization is necessary: a model trained on punctuated text reaches D1 macro-F1 0.959 on punctuated text and collapses to 0.381 on ASR-style text.
- Replay demo (illustrative, not an evaluation): a naive agent acted as wanted at 3 of 17 scripted points, floorcall at 14 of 18.

1. The decisions

Decisions fire on pipeline events, and each event asks only its own questions, in one pass. The policy turns the calibrated probabilities into actions through thresholds chosen on calibration data, each a point on a measured tradeoff curve.

Event	Questions in one pass
user_pause	turn_complete, route, escalate
user_speech_during_agent	barge_in, escalate

- turn_complete: Has the user finished their turn and is now waiting for a reply? Options: false, true.
- barge_in: The user spoke while the agent was talking. What was it? Options: backchannel, interruption, noise.
- route: Which banking request is the user making? Options: the 15 banking intents of CLINC150 and out_of_scope.
- escalate: Should this conversation be handed to a human? Options: false, true.

Why not an LLM: the inline decisions must fit in the tens of milliseconds a voice agent has left after speech recognition, the response model and speech synthesis. A prompted LLM called over the network measured p50 2,565 ms in Table B.

2. Data

Decision	Source	Licence	Train	Calib	Test	Test balance
D1 turn_complete	SwDA	CC BY-NC-SA 3.0	49,381	8,201	14,998	false 8,843, true 6,155
D2 barge_in	SwDA	CC BY-NC-SA 3.0	44,187	7,407	13,360	backchannel 5,659, interruption 6,700, noise 1,001
D3 route	CLINC150	CC BY 3.0	1,750	400	1,449	16 labels
D4 escalate	Twitter support	CC BY-NC-SA 4.0	6,000	100	200	false 111, true 89

- SwDA is split by conversation, and D1 and D2 share the split, so no test conversation is seen through either task. Test sets were frozen and hashed before the first training run, and every result records the hash it was scored on.
- All text, in training and at inference, is normalized as a speech recognizer delivers it: lowercase, punctuation removed.
- D4's test and calib labels are one person's, by hand, on real customer messages; its training labels come from an LLM (gpt-oss-120b). Relabelling blind under a revised guideline, the labeller agreed with the first pass at accuracy 0.783, Cohen's κ 0.551 ($n = 300$); the LLM labels agreed with the hand labels on calib at κ 0.420, with escalate recall 0.426.
- D4's messages are real tweets, some with personal details, so the public repository holds their ids, labels and hashes, not their text; `floorcall data restore-escalate` rebuilds them byte for byte from the corpus.

3. Model and training

Base model: [convainnovations/laya](#) (English), revision 55cf4c4, a ModernBERT-large encoder with a decision head. One multi-task checkpoint is fine-tuned on all four decisions with soft cross-entropy; the kept epoch is the one with the lowest cross-entropy on a dev split held out of train by conversation. Temperatures are then fitted per decision on calib (D1 1.08, D2 1.20, D3 1.92, D4 4.86), and D4's decision threshold is chosen on calib by macro-F1. The fine-tuned rows below are this checkpoint, released as [enz23/floorcall](#) under CC BY-NC-SA 4.0.

4. Results

Table A: quality per decision. Brackets are 95% intervals.

D1 turn_complete

Model	Accuracy [95% CI]	Macro-F1	ECE	Brier	Hard-subset acc. (n)
majority class (train prior)	0.590 [0.582, 0.597]	0.371	0.000	0.484	1.000 (775)
TF-IDF + logistic regression	0.624 [0.616, 0.632]	0.544 [0.536, 0.553]	0.008 [0.006, 0.018]	0.456 [0.452, 0.460]	0.843 [0.817, 0.868] (775)
stock Laya, zero-shot	0.488 [0.480, 0.496]	0.485	0.028	0.510	0.505 (775)
fine-tuned	0.820 [0.813, 0.826]	0.813 [0.807, 0.820]	0.020 [0.015, 0.026]	0.243 [0.237, 0.250]	0.725 [0.693, 0.755] (775)
fine-tuned + temperature	0.820 [0.813, 0.826]	0.813 [0.807, 0.820]	0.010 [0.007, 0.016]	0.243 [0.236, 0.249]	0.725 [0.693, 0.755] (775)
prompted LLM, stated probabilities	0.721 [0.714, 0.728]	0.721	0.138	0.435	0.412 (775)

D2 barge_in

Model	Accuracy [95% CI]	Macro-F1	ECE	Brier	Hard-subset acc. (n)
majority class (train prior)	0.501 [0.493, 0.510]	0.223	0.010	0.564	0.453 (5012)
lexical rule: backchannel words + length	0.951 [0.948, 0.955]	0.908 [0.901, 0.916]	0.001 [0.000, 0.005]	0.092 [0.085, 0.098]	0.985 [0.981, 0.988] (5012)
TF-IDF + logistic regression	0.931 [0.926, 0.935]	0.850 [0.840, 0.859]	0.015 [0.012, 0.019]	0.102 [0.097, 0.107]	0.963 [0.958, 0.968] (5012)
stock Laya, zero-shot	0.368 [0.360, 0.376]	0.238	0.005	0.661	0.439 (5012)

Model	Accuracy [95% CI]	Macro-F1	ECE	Brier	Hard-subset acc. (n)
fine-tuned	0.969 [0.966, 0.972]	0.937 [0.931, 0.943]	0.008 [0.006, 0.010]	0.042 [0.039, 0.046]	0.977 [0.972, 0.981] (5012)
fine-tuned + temperature	0.969 [0.966, 0.972]	0.937 [0.931, 0.943]	0.003 [0.003, 0.006]	0.042 [0.038, 0.045]	0.977 [0.972, 0.981] (5012)
prompted LLM, stated probabilities	0.708 [0.700, 0.716]	0.574	0.196	0.493	0.711 (5012)

D3 route

Model	Accuracy [95% CI]	Macro-F1	ECE	Brier	Hard-subset acc. (n)
majority class (train prior)	0.690 [0.666, 0.713]	0.051	0.547	0.837	n/a
TF-IDF + logistic regression	0.908 [0.893, 0.923]	0.848 [0.820, 0.872]	0.057 [0.046, 0.071]	0.147 [0.130, 0.165]	n/a
stock Laya, zero-shot	0.934 [0.921, 0.946]	0.866	0.030	0.115	n/a
fine-tuned	0.948 [0.937, 0.959]	0.906 [0.881, 0.926]	0.043 [0.033, 0.054]	0.090 [0.071, 0.111]	n/a
fine-tuned + temperature	0.948 [0.937, 0.959]	0.906 [0.881, 0.926]	0.011 [0.008, 0.023]	0.082 [0.065, 0.100]	n/a
prompted LLM, stated probabilities	0.970 [0.961, 0.979]	0.939 [0.918, 0.956]	0.028 [0.021, 0.037]	0.058 [0.044, 0.074]	n/a

D4 escalate

Model	Accuracy [95% CI]	Macro-F1	ECE	Brier	Hard-subset acc. (n)
majority class (calib prior)	0.555 [0.485, 0.625]	0.357 [0.327, 0.385]	0.025 [0.000, 0.095]	0.495 [0.487, 0.504]	0.710 [0.620, 0.800] (100)
TF-IDF + logistic regression ($\theta = 0.240$)	0.685 [0.620, 0.750]	0.681 [0.613, 0.743]	0.167 [0.125, 0.239]	0.473 [0.386, 0.561]	0.740 [0.650, 0.820] (100)
stock Laya, zero-shot	0.665 [0.600, 0.730]	0.625 [0.554, 0.693]	0.092 [0.057, 0.159]	0.422 [0.396, 0.447]	0.730 [0.640, 0.810] (100)
stock Laya, calib threshold ($\theta = 0.505$)	0.645 [0.580, 0.710]	0.579 [0.506, 0.649]	0.092 [0.057, 0.159]	0.422 [0.396, 0.447]	0.700 [0.610, 0.790] (100)
fine-tuned	0.695 [0.630, 0.755]	0.672 [0.602, 0.737]	0.277 [0.220, 0.340]	0.544 [0.434, 0.658]	0.740 [0.650, 0.820] (100)
fine-tuned + temperature	0.695 [0.630, 0.755]	0.672 [0.602, 0.737]	0.064 [0.040, 0.135]	0.418 [0.363, 0.475]	0.740 [0.650, 0.820] (100)
fine-tuned + temperature, calib threshold ($\theta = 0.505$)	0.705 [0.640, 0.765]	0.681 [0.612, 0.747]	0.064 [0.040, 0.135]	0.418 [0.363, 0.475]	0.750 [0.660, 0.830] (100)
prompted LLM, stated probabilities	0.635 [0.570, 0.700]	0.610 [0.538, 0.677]	0.248 [0.186, 0.312]	0.539 [0.445, 0.635]	0.570 [0.470, 0.660] (100)

Table B: latency at batch 1. Budgets: p99 at most 50 ms on GPU, 100 ms on CPU.

Path	p50 ms	p95 ms	p99 ms	of which forward, p50	of which packing, p50	Fits budget (p99)
GPU, CUDA graphs: user_pause, 3 questions in 1 call	29.1	55.6	59.0	24.6	0.8	no (≤ 50 ms)
GPU, CUDA graphs: user_pause, 3 questions in 3 calls	52.4	79.2	85.2	43.1	0.8	no (≤ 50 ms)
GPU, CUDA graphs: user_speech_during_agent, 2 questions in 1 call	44.3	54.9	57.4	38.3	1.9	no (≤ 50 ms)
GPU, eager: user_pause, 3 questions in 1 call	56.4	82.3	88.5	50.3	1.1	no (≤ 50 ms)
GPU, eager: user_pause, 3 questions in 3 calls	120.4	158.6	179.3	110.0	0.9	no (≤ 50 ms)
GPU, eager: user_speech_during_agent, 2 questions in 1 call	55.4	66.4	73.7	49.1	2.5	no (≤ 50 ms)
CPU: user_pause, 3 questions in 1 call	976.9	2162.4	2219.8	973.1	0.7	no (≤ 100 ms)
CPU: user_pause, 3 questions in 3 calls	680.9	1899.1	1953.6	674.0	0.6	no (≤ 100 ms)
CPU: user_speech_during_agent, 2 questions in 1 call	1153.6	1426.7	1464.5	1149.7	1.4	no (≤ 100 ms)

Path	p50 ms	p95 ms	p99 ms	of which forward, p50	of which packing, p50	Fits budget (p99)
prompted LLM (gpt-oss-20b via OpenRouter, pinned DeepInfra bf16), user_pause, 3 questions in 1 call: network latency, request to parsed answer	2565.1	4270.7	6072.2	n/a	n/a	no (≤ 50 ms)

Measured on NVIDIA GeForce RTX 5070 Ti Laptop GPU (driver 591.86) and Intel64 Family 6 Model 197 Stepping 2, GenuineIntel with 16 torch threads; on AC power: True; torch 2.14.0+cu130, laya 0.3.21. GPU rows measured 2026-09-30 (UTC). Windows power mode: Best performance. GPU power limit: vendor default, 80 W base plus Dynamic Boost (115 W enforced at the start). Batch 1, 50 warmup and 1000 timed iterations over 50 fixed inputs per event; every timed call is a full `Decider.decide` (packing, tokenizing, forward, temperatures).

Table C: robustness to ASR-style noise. Cells are macro-F1 (accuracy).

Decision	noise 0.00	noise 0.05	noise 0.10	noise 0.20
D1 turn_complete	0.814 (0.820)	0.775 (0.786)	0.743 (0.760)	0.679 (0.713)
D2 barge_in	0.937 (0.969)	0.865 (0.927)	0.810 (0.887)	0.728 (0.814)
D3 route	0.906 (0.948)	0.882 (0.937)	0.845 (0.919)	0.759 (0.878)
D4 escalate	0.672 (0.695)	0.650 (0.675)	0.643 (0.680)	0.631 (0.665)

Table D: ablations. Each arm is trained again without part of the state, on Kaggle, and compared with a full arm trained the same way there. Each arm is one run; differences smaller than the gap between two runs of one recipe are not read as effects.

Variant	D1 macro-F1 (acc)	D1 hard acc.	D2 macro-F1 (acc)	D2 hard acc.	D3 macro-F1 (acc)	D4 macro-F1 (acc)
full model	0.824 (0.827)	0.675	0.941 (0.973)	0.994	0.911 (0.955)	0.695 (0.715)
without recent_turns	0.823 (0.826)	0.717	0.941 (0.971)	0.982	0.924 (0.958)	0.629 (0.680)
without agent_last_utterance	0.820 (0.823)	0.720	0.942 (0.971)	0.987	0.929 (0.964)	0.648 (0.685)
without normalization, scored on written text	0.959 (0.960)	0.992	0.945 (0.972)	0.981	0.925 (0.957)	0.695 (0.715)
without normalization, scored on ASR-style text	0.381 (0.593)	1.000	0.938 (0.969)	0.979	0.927 (0.959)	0.701 (0.720)

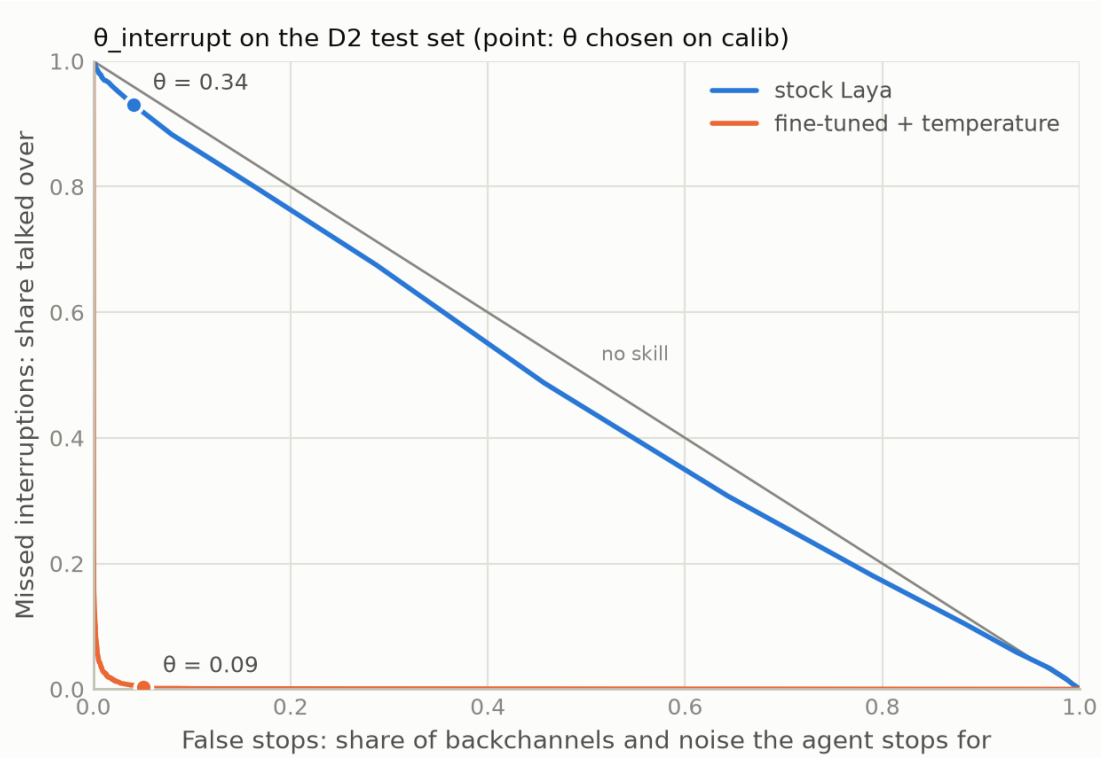
Each variant minus the full arm, with paired-bootstrap 95% intervals:

Variant minus full arm	D1 macro-F1	D1 hard acc.	D2 macro-F1	D2 hard acc.	D3 macro-F1	D4 macro-F1
without recent_turns	-0.001 [-0.006, +0.003]	+0.043 [+0.021, +0.066]	-0.000 [-0.007, +0.005]	-0.012 [-0.015, -0.009]	+0.013 [-0.005, +0.032]	-0.066 [-0.130, -0.004]
without agent_last_utterance	-0.004 [-0.009, -0.000]	+0.045 [+0.023, +0.067]	+0.000 [-0.005, +0.006]	-0.008 [-0.011, -0.005]	+0.018 [+0.000, +0.037]	-0.047 [-0.101, +0.004]
without normalization, scored on written text	+0.134 [+0.128, +0.141]	+0.317 [+0.285, +0.350]	+0.004 [-0.002, +0.010]	-0.014 [-0.017, -0.010]	+0.015 [-0.004, +0.034]	+0.000 [-0.045, +0.046]
without normalization, scored on ASR-style text	-0.443 [-0.451, -0.436]	+0.325 [+0.293, +0.359]	-0.003 [-0.009, +0.002]	-0.015 [-0.019, -0.012]	+0.017 [-0.001, +0.036]	+0.006 [-0.039, +0.052]

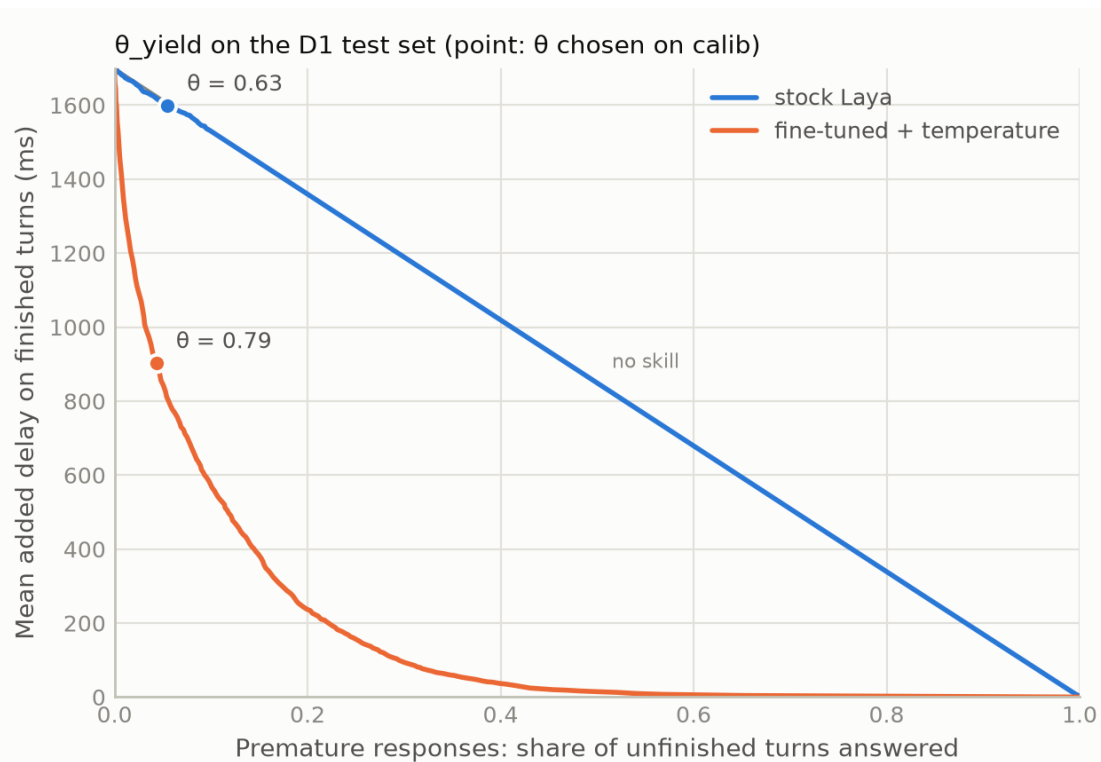
Operating points, each threshold chosen on calib:

Model	$\theta_{\text{interrupt}}$	false stops (test)	missed interruptions (test)	θ_{yield}	premature responses (test)	added delay, ms (test)
stock Laya	0.340	0.040	0.931	0.630	0.055	1599
fine-tuned + temperature	0.090	0.050	0.003	0.785	0.043	903

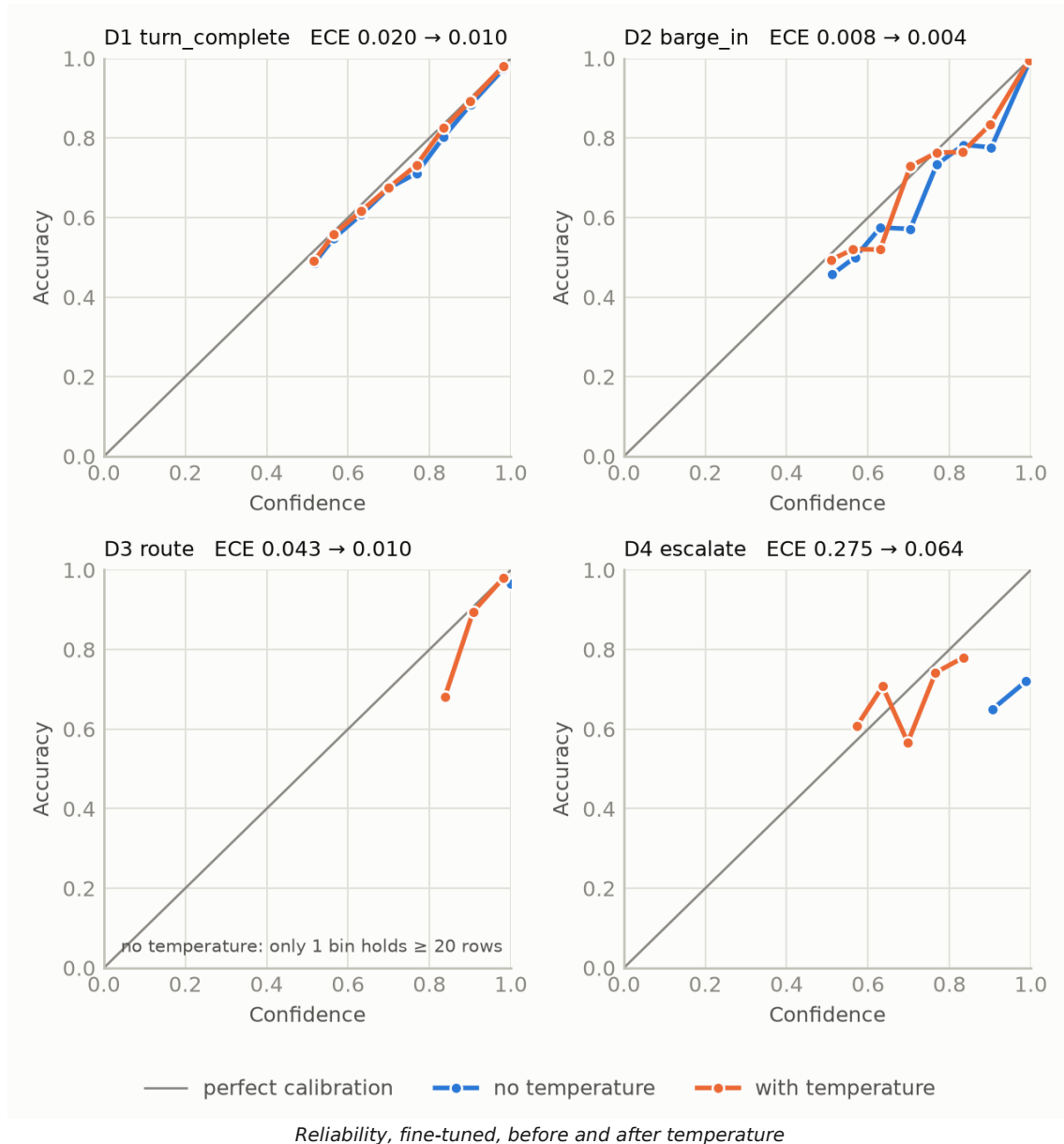
The curves and reliability figures are computed at fp16 inference, the default since D-040; Table A's fine-tuned rows were scored before that change, so a figure's ECE can differ from Table A's in the third decimal.



$\theta_{\text{interrupt}}$: false stops against missed interruptions



θ_{yield} : premature responses against added delay



5. Replay: the same call through two agents

Eight scripted banking calls run through a naive agent, which answers after 800 ms of silence and stops for any speech, and through floorcall. The scripts were frozen by hash before floorcall first ran on them. The naive agent acted as wanted at 3 of 17 points and floorcall at 14 of 18. floorcall's misses:

- 02_real_interruption: wanted respond, route report_fraud; floorcall: respond, route transactions ($p(\text{turn_complete})=0.83 \geq 0.785$).
- 04_side_talk: wanted ignore; floorcall: stop and listen ($p(\text{interruption})=0.73 \geq 0.09$).
- 08_escalation: wanted keep listening; floorcall: respond, route spending_history ($p(\text{turn_complete})=0.88 \geq 0.785$).
- 08_escalation: wanted stop and listen, hand off; floorcall: stop and listen ($p(\text{interruption})=0.74 \geq 0.09$).

6. Limitations

- Latency: no measured GPU path meets $p99 \leq 50$ ms on the laptop GPU, and every CPU path misses $p99 \leq 100$ ms by a wide margin (Table B).
- D2: a lexical rule on backchannel words and length reaches macro-F1 0.908 against the model's 0.937, and is better on D2's hard subset (0.985 against 0.977).
- D4: its labels have a low ceiling (section 2), its test set is 200 messages, and at its calib threshold it ties

TF-IDF with logistic regression on macro-F1 (+0.000 [-0.061, +0.060]); it is better calibrated.

- D1 reads text only and cannot hear intonation; SwDA is two people on the phone from 1990 to 1991, not a user talking to an agent. Its hard subset, truncations ending on a turn-final word, is right at 0.725.
- ASR noise costs accuracy everywhere (Table C): D2 falls from 0.937 to 0.728 macro-F1 at the heaviest noise level.
- Table D's arms are single runs, so its hard-subset movements cannot be separated from run-to-run variation.

7. Reproducing it

```
git clone https://github.com/yashraz23/floorcall
cd floorcall
uv sync                                # NVIDIA GPU
uv run pytest
uv run floorcall data restore-escalate  # D4 text, from the corpus
uv run floorcall replay demo/scripts/*.json --compare naive
uv run floorcall eval readme           # every table, from results/
uv run floorcall release report
```

- Every result file records the commit that produced it. The public history was rewritten once to remove D4's tweet text; docs/commit-map.txt maps the commits cited by results files to their public SHAs.
- Every divergence from the original plan, and why, is in docs/DECISIONS.md.
- Licences: code Apache-2.0; the weights CC BY-NC-SA 4.0; data files carry their sources' licences (data/LICENSE.md): SwDA CC BY-NC-SA 3.0, CLINC150 CC BY 3.0, Customer Support on Twitter CC BY-NC-SA 4.0.